

Ex.No. 3:

AIM: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss.

Objective:

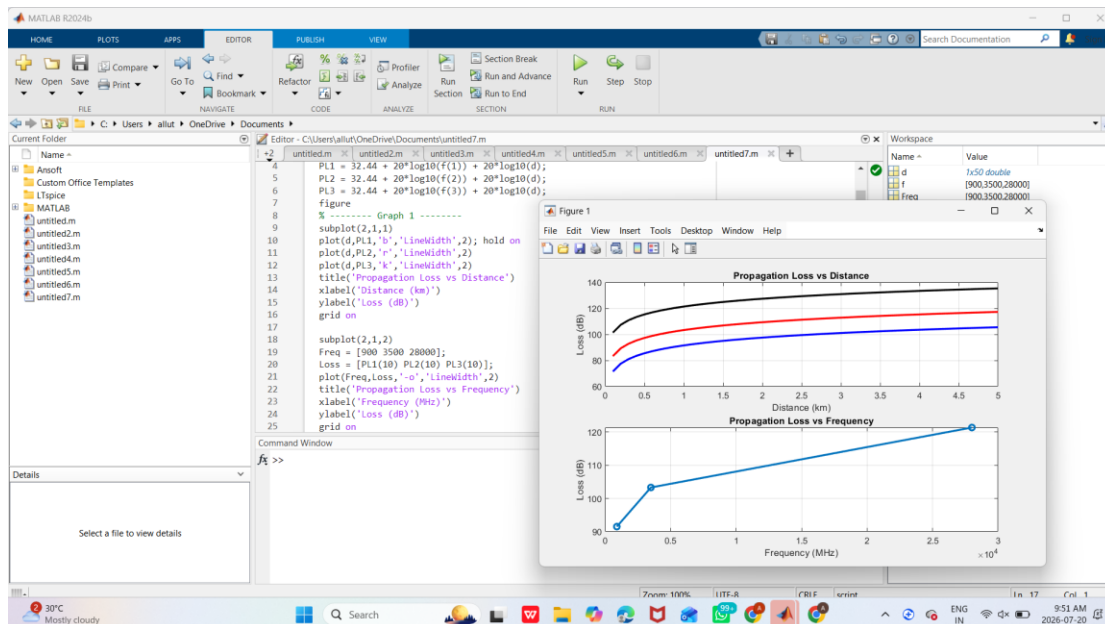
1. To compute the **Propagation Loss (dB)** for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models.

Objective 1: Matlab Code and Output

```
clc; clear; close all;
d = 0.1:0.1:5;
f = [900 3500 28000];
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
subplot(2,1,1)plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on

subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```

OUTPUT:



Objective 2:

To evaluate the impact of different **Operating Frequencies (GHz)** on signal attenuation and coverage performance.

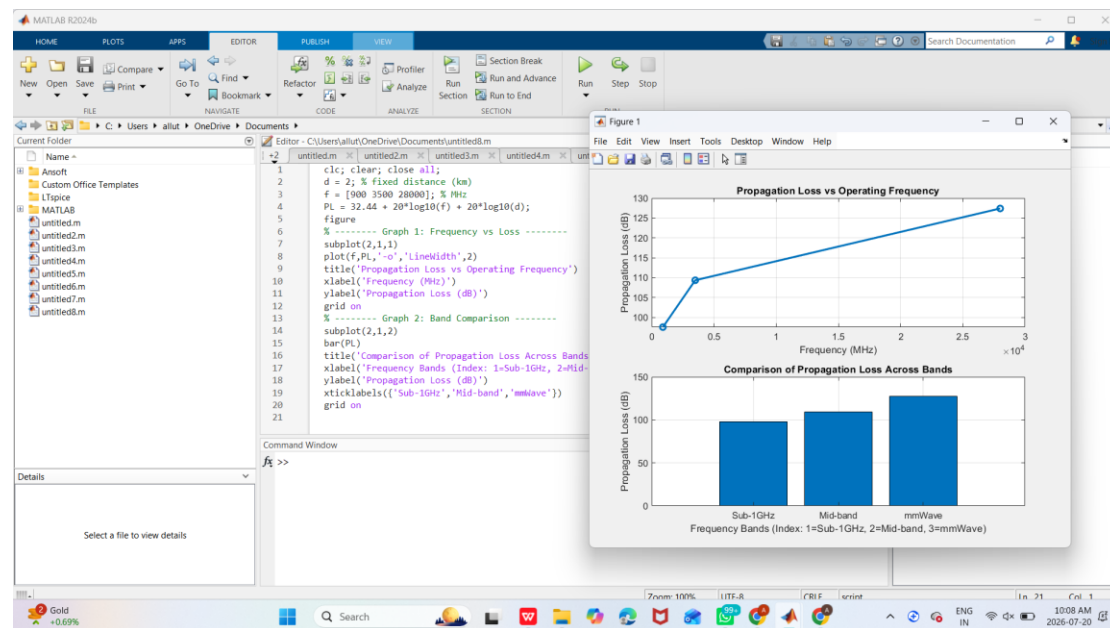
Matlab Code and Output :

```

clc; clear; close all;
d = 2; f = [900 3500 28000];
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
subplot(2,1,1)
plot(f,PL,'-o','LineWidth',2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
subplot(2,1,2)
bar(PL)
title('Comparison of Propagation Loss Across Bands')
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
grid on

```

OUTPUT:



Objective 3:

To analyze how **Communication Distance (km)** affects propagation loss across different spectrum bands.

Objective 3: Matlab Code and Output

```
clc; clear; close all;  
f = [900 3500 28000];  
d = 2;
```

```
PL = 32.44 + 20*log10(f) + 20*log10(d);  
figure
```

```
subplot(2,1,1)  
plot(f, PL, '-o', 'LineWidth', 2)  
title('Propagation Loss vs Operating Frequency')  
xlabel('Frequency (MHz)')  
ylabel('Propagation Loss (dB)')  
grid on  
subplot(2,1,2)
```

```

plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
grid on

```

OUTPUT:

